

Statistical Learning - W09

Setup: Multi-Class Classification

Class Structure

Consider the following class setup:

Class	Distribution	Sample Size
Class 1	$N_{25}(\mu_1, \Sigma_1)$	$n_1 = 20$
Class 2	$N_{25}(\mu_2, \Sigma_2)$	$n_2 = 20$
$\vdots$	$\vdots$	$\vdots$
Class 5	$N_{25}(\mu_5, \Sigma_5)$	$n_5 = 20$

Observations for Class 1: $X_{11}, \dots, X_{120}$.

Estimating the Covariance Matrix

The sample covariance for Class 1 is:

$$\hat{\Sigma}_1 = \frac{1}{20} \sum_{i=1}^{20} (X_{1i} - \bar{X}_1)(X_{1i} - \bar{X}_1)'$$

Question: Can I compute $\hat{\Sigma}_1^{-1}$?

Note the following rank fact:

$$\text{rank}(\hat{\Sigma}_1) \text{ (} 25 \times 25 \text{ matrix)}$$

$\hat{\Sigma}_1^{-1}$ **does not exist.**

Also recall: for a matrix A and vectors,

$$\begin{matrix} A & A' & A'A & AA' \\ (X_i - \bar{X}_1)(X_i - \bar{X}_1)' = (X_i - \bar{X}_1) \end{matrix}$$

(outer product, rank-1 matrix)

Also: $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.

$$\sum (X_i - \bar{X}_i) = 0$$

The pooled sample covariance:

$$\hat{\Sigma} = \frac{\sum_{i=1}^5 \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)(X_{ij} - \bar{X}_i)'}{n_1 + \dots + n_5}$$

Regularization for Singular Covariance Matrices

Problem Motivation

In such cases, **one option** is to use *generalized inverses* of $\hat{\Sigma}_1, \dots, \hat{\Sigma}_5$.

Other option is to use regularization, where we can use:

$$\hat{\Sigma}_{i,\text{reg}} = \hat{\Sigma}_i + \lambda I \quad \text{for } i = 1, 2, \dots, 5$$

or

$$\hat{\Sigma}_{i,\text{reg}} = \hat{\Sigma}_i + \lambda \hat{\Sigma} \quad \text{for } i = 1, 2, \dots, 5$$

where λ is usually chosen by cross-validation.

This method is known as **regularized discriminant analysis**.

For QDA, when the $\hat{\Sigma}_i$'s are not invertible, we use one of these regularizations.

When Total Observations < Dimension

Class	dim./obs.
1	30
2	40
$\vdots$	20
$\vdots$	10
5	58
Total	150

Note that if the **total number of observations** < **dim. of the data**, $\hat{\Sigma}$ becomes a singular matrix. So, we **cannot** use $\hat{\Sigma}^{-1}$ for LDA.

$\hat{\Sigma}_1, \hat{\Sigma}_2, \dots, \hat{\Sigma}_5$ are singular. $\hat{\Sigma}$ is also singular.

5 classes	$\hat{\Sigma}_i$
n_1	$n_1 - 1$
n_2	$n_2 - 1$
n_3	$n_3 - 1$
n_4	$n_4 - 1$
n_5	$n_5 - 1$

Regularization for LDA

In such cases, for LDA, we use:

$$\hat{\Sigma}_{\text{reg}} = \hat{\Sigma} + \lambda I \quad \text{for some appropriate choice of } \lambda \text{ (usually chosen by CV)}$$

Regularization for QDA

For QDA:

$$\hat{\Sigma}_{i,\text{reg}} = \left(\hat{\Sigma}_i \right) + \lambda \hat{\Sigma} \quad \text{may not work.}$$

However, we can use:

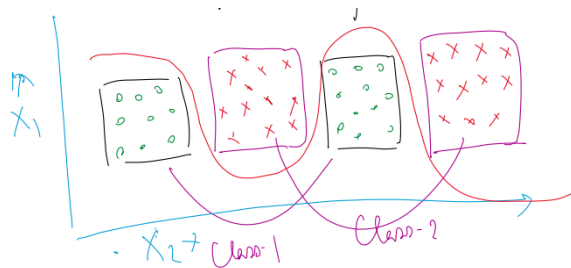
$$\hat{\Sigma}_{i,\text{reg}} = \hat{\Sigma}_i + \lambda I \quad \text{or}$$

$$\hat{\Sigma}_{i,\text{reg}} = \hat{\Sigma}_i + \lambda \left(\alpha \hat{\Sigma} + (1 - \alpha) I \right)$$

$$\approx \hat{\Sigma}_i + \lambda_1 \hat{\Sigma} + \lambda_2 I, \quad \text{where } \lambda_1, \lambda_2 \text{ are to be chosen from the data.}$$

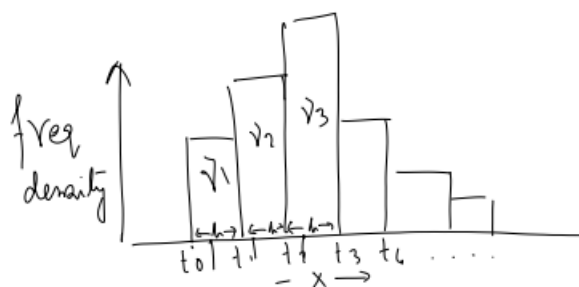
Also note: $\hat{\Sigma} \approx n^{-1} \delta$.

Classification Based on Nonparametric Density Estimates



Histogram Density Estimate

Classification based on nonparametric density estimates



The histogram density estimate is:

$$\hat{f}_H(x) = \frac{\nu_i/h}{n} \quad t_{i-1} \leq x < t_i$$

Histogram density estimate for $f(x)$:

Given $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f$,

$$\hat{f}_H(x) = \frac{\nu_i/h}{n}$$

where $\nu_i = \# \text{obs. in } [t_{i-1}, t_i]$.

Does $\hat{f}_H(x) \xrightarrow{p} f(x)$?

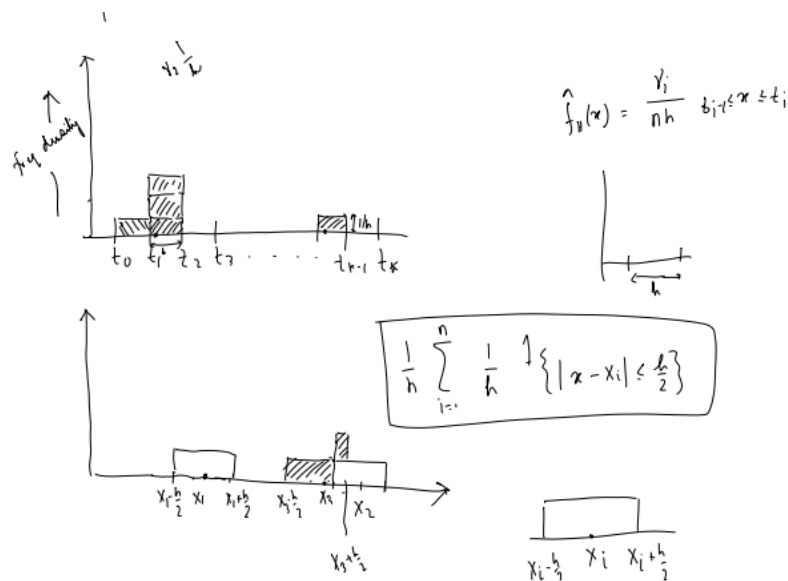
$$\nu_i \sim \text{Bin}\left(n, p_i = \int_{t_{i-1}}^{t_i} f(x) dx\right)$$

Drawbacks of the histogram density estimate:

- Depends on the choice of the class boundaries.
- It is not continuous at all; the derivative is not meaningful.
- Depends on the choice of h ; one needs to choose it appropriately.
- Difficult to use it for high-dimensional problems.

We write the histogram density estimate with bandwidth h as:

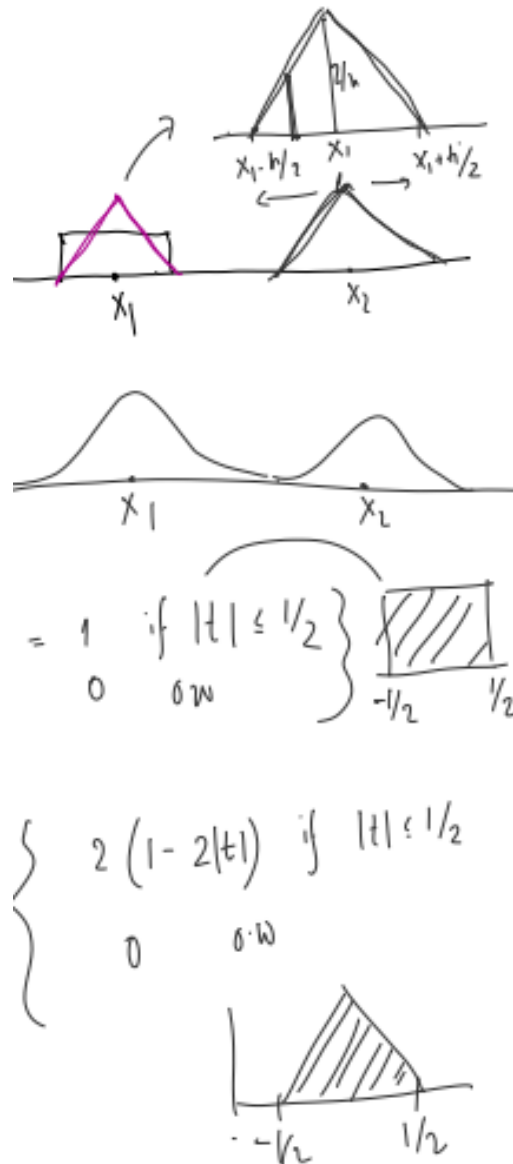
$$\hat{f}_H(x) = \frac{\nu_i}{nh}, \quad t_{i-1} \leq x \leq t_i$$



Kernel Density Estimation (Univariate)

Consider a moving window centred at x with half-width $h/2$:

$$\frac{1}{h} \sum_{i=1}^n \frac{1}{h} \mathbf{1}_{\{|x - X_i| \leq \frac{h}{2}\}}$$



The flat (uniform) kernel estimate is:

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} \mathbf{1}\{|x - X_i| \leq \frac{h}{2}\}$$

General Kernel Density Estimator

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} \mathbf{1}\{|x - X_i| \leq h/2\} = \frac{1}{n} \sum_{i=1}^n \frac{2}{h} \left(1 - \frac{|x - X_i|}{h/2}\right) \mathbf{1}\{|x - X_i| \leq h/2\}$$

The second form above is the **triangular kernel**:

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x - X_i}{h}\right) \quad \text{where } K(t) = \begin{cases} 1 & \text{if } |t| \leq 1/2 \\ 0 & \text{o.w.} \end{cases}$$

Uniform (rectangular) kernel: $K(t) = 1$ for $|t| \leq 1/2$, 0 otherwise.

Similarly:

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n \frac{2}{h} \left(1 - \frac{|x - X_i|}{h/2} \right) \mathbf{1}\{|x - X_i| \leq h/2\} = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x - X_i}{h}\right)$$

where

$$K(t) = \begin{cases} 2(1 - 2|t|) & \text{if } |t| \leq 1/2 \\ 0 & \text{o.w.} \end{cases}$$

Triangular kernel shape on $[-1/2, 1/2]$.

General Kernel Density Estimate:

In general, we can consider a density function of the form

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x - X_i}{h}\right),$$

where $K(\cdot)$ is a **univariate density function**.

This is known as the **Kernel Density Estimate**.

Gaussian Kernel

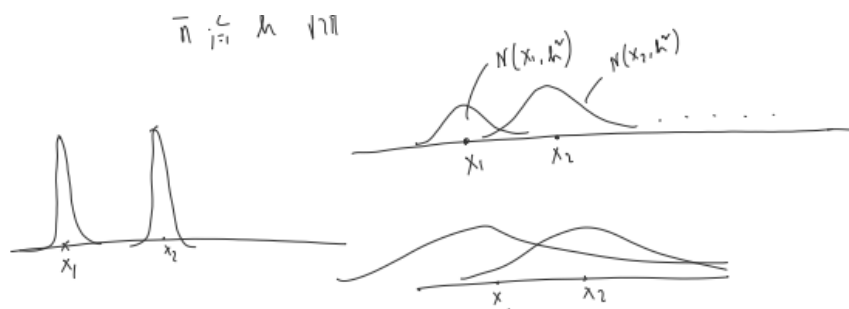
If

$$K(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}t^2},$$

a Gaussian kernel, the density estimate turns out to be:

$$\frac{1}{n} \sum_{i=1}^n \frac{1}{h} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x - X_i}{h}\right)^2}$$

which is an average of $N(X_i, h^2)$ densities.



Larger bandwidth: smoother combined density estimate $\hat{f}_n(x)$.

Here h **controls the smoothness** of the density estimates. Larger the value of h , smoother the density estimates. This smoothing parameter h is also known as the **bandwidth**.

Also recall:

$$\text{MSE} = E\left(\hat{\theta} - \theta\right)^2$$

Mean Integrated Squared Error (MISE)

$$\begin{aligned}\text{MISE} &= \int E\left(\hat{f}_h(x) - f(x)\right)^2 dx \\ \text{IMSE} &= E \int \left(\hat{f}_h(x) - f(x)\right)^2 dx\end{aligned}$$

Also,

$$\begin{aligned}\text{ISE} &= \int \left(\hat{f}_h(x) - f(x)\right)^2 dx \\ &= \int \hat{f}_h^2(x) dx + \int f^2(x) dx - 2 \int \hat{f}_h(x) f(x) dx \\ &\quad \Downarrow \\ &\quad -2 E\left(\hat{f}_h(x)\right)\end{aligned}$$

Kernel Density Estimation

Kernel Density Estimation

If $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} f$, then the kernel density estimator of $f(x)$ is given by:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right),$$

where K is a kernel function (usually a one-dimensional density function) and h is the associated smoothing parameter known as the **bandwidth**.

Note: $\hat{f}_h(x)$ can be viewed as an average of i.i.d. random variables. So, suitable laws of large numbers and limit theorems are applicable.

Bias and Variance of KDE

One can show that:

$$E \hat{f}_h(x) = f(x) + O(h^2) \quad \text{if the kernel is a symmetric kernel}$$

$$\text{Var } \hat{f}_h(x) = O\left(\frac{1}{nh}\right)$$

If $h \rightarrow 0$ and $nh \rightarrow \infty$ as $n \rightarrow \infty$, then $\hat{f}_h(x) \xrightarrow{p} f(x)$.

MSE of KDE

$$\text{MSE} = E\left(\hat{f}_h(x) - f(x)\right)^2 = c_1 h^4 + c_2 \frac{1}{nh} =: \Psi(h)$$

$$\frac{\partial \Psi(h)}{\partial h} = 0 \implies h \propto n^{-1/5}$$

$$\left[4c_1 h^3 - c_2 \frac{1}{nh^2} = 0 \implies h : O(n^{-1/5}) \right]$$

$$\text{MSE}_{\text{opt}} = c_1^* n^{-4/5} + c_2^* n^{-4/5} = O(n^{-4/5})$$

$$\sqrt{n}(\hat{\theta} - \theta) \rightarrow \dots$$

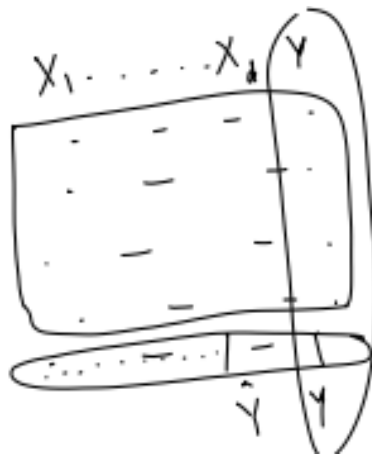
$$E(\hat{\theta} - \theta)^2 \approx \frac{1}{n} : O(n^{-1})$$

$$O(n^{-1}) \quad O(n^{-4/5})$$

Even for MISE, we get the same order for opt h and opt MISE.

Choosing the Bandwidth in Practice

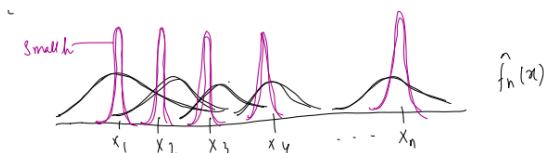
- How to choose the value of h in practice?



$$\text{MISE} = E \int \left(\hat{f}_h(x) - f(x) \right)^2 dx = \int E \left(\hat{f}_h(x) - f(x) \right)^2 dx = \text{IMSE}$$

$$\text{ISE} = \int \left(\hat{f}_h(x) - f(x) \right)^2 dx = \int \hat{f}_h^2(x) dx + \int f^2(x) dx - 2 \int \hat{f}_h(x) f(x) dx$$

ISE can serve as a proxy for MISE, but ISE cannot be calculated since it involves unknown f .



Note that the **first term** does not involve f , and it can be calculated. The **second term** does not involve h , and can be dropped. The **third term** can be estimated by:

$$\int \hat{f}_h(x) f(x) dx = E(\hat{f}_h(x)) \cong \frac{1}{n} \sum_{i=1}^n \hat{f}_h(X_i)$$

Small h : many narrow Gaussian bumps (overfitting), vs. combined $\hat{f}_n(x)$ (black curve). We estimate $\int \hat{f}_h(x) f(x) dx$ by

$$\frac{1}{n} \sum_{i=1}^n \hat{f}_{h,-i}(X_i) \quad \leftarrow \begin{cases} n\text{-fold CV} \\ \text{or leave-one-out CV} \end{cases}$$

where $\hat{f}_{h,-i}$ is the density estimate based on $n - 1$ observations where X_i is left out.

So, one minimizes:

$$\int \hat{f}_h^2(x) dx - \frac{2}{n} \sum_{i=1}^n \hat{f}_{h,-i}(X_i)$$

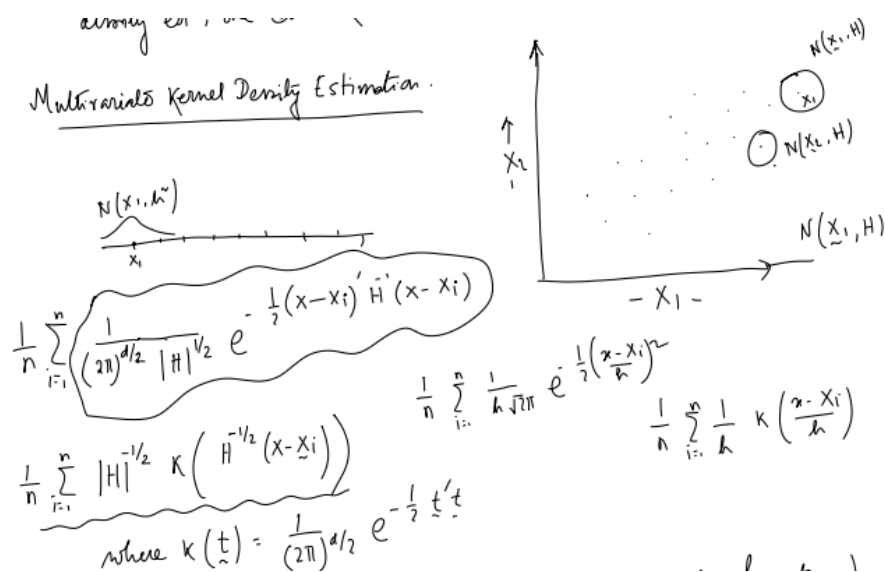
where

$$\hat{f}_{h,-i}(X_i) = \frac{1}{n-1} \sum_{\substack{j=1 \\ j \neq i}}^n \frac{1}{h} K\left(\frac{X_i - X_j}{h}\right)$$

This method of choosing the bandwidth is known as **Least Square Cross Validation**.

For other methods of choosing bandwidths for density estimation, one can look at *Silverman (1986)*, *Scott (1992)*.

Multivariate Kernel Density Estimation



The multivariate KDE with bandwidth matrix H is:

$$\frac{1}{n} \sum_{i=1}^n \left(\frac{1}{(2\pi)^{d/2} |H|^{1/2}} \right) e^{-\frac{1}{2}(x-X_i)'H^{-1}(x-X_i)}$$

This can be written as:

$$\frac{1}{n} \sum_{i=1}^n \frac{1}{h\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-X_i}{h}\right)^2} \quad \text{or} \quad \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x-X_i}{h}\right)$$

$$\frac{1}{n} \sum_{i=1}^n |H|^{-1/2} K(H^{-1/2}(x - X_i))$$

where $K(t) = \frac{1}{(2\pi)^{d/2}} e^{-\frac{1}{2}t't}$.

If $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} f$ (where f is a d -dimensional density function), the **multivariate kernel density estimator** of $f(x)$ is given by:

$$\hat{f}_H(x) = \frac{1}{n} \sum_{i=1}^n |H|^{-1/2} K(H^{-1/2}(x - X_i)),$$

where $K(\cdot)$ is a d -dimensional kernel function (usually a d -dimensional density function) and H is the bandwidth matrix.

$$H = \begin{bmatrix} \lambda & & \\ & \ddots & \\ & & \lambda \end{bmatrix} = CC'$$

Instead of working with a general bandwidth matrix, it is computationally more convenient to **standardize the data** and use $H = h^2 I$ for density estimation.

Standardization Approach

Suppose that $Y_1, Y_2, \dots, Y_n$ are the standardized observations. Then $\hat{f}_h(y)$ is given by:

$$\hat{f}_h(y) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h^d} K\left(\frac{y - Y_i}{h}\right)$$

$$E\left(\hat{f}_h(y)\right) = f(y) + O(h^2) \quad [\text{Scott (1992)}]$$

$$\text{Var}\left(\hat{f}_h(x)\right) = O\left(\frac{1}{nh^d}\right)$$

If $h \rightarrow 0$ and $nh^d \rightarrow \infty$ as $n \rightarrow \infty$, then $\hat{f}_h(y) \xrightarrow{p} f(y)$.

$$\begin{aligned} \text{Opt } h &= O\left(n^{-\frac{1}{d+4}}\right) & \text{Opt MSE} &= O\left(n^{-\frac{4}{d+4}}\right) \\ &O(n^{-1}) & &O(n^{-4/5}) \end{aligned}$$

For MISE, the orders remain the same.

Data and the Standardized Variables

Given $X_1, \dots, X_n$: d -dimensional observations,

$\hat{\Sigma}$: estimated dispersion matrix

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})'$$

$$Y_j = \hat{\Sigma}^{-1/2}(X_j - \bar{X}) \quad \text{for } j = 1, 2, \dots, n$$

Two Options for KDE in Classification (KDA)

Option 1

Given $X_1, \dots, X_n$: d -dimensional observations and a query point,

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h^d} K\left(\frac{x - X_i}{h}\right)$$

Option 2

$$Y_i = \hat{\Sigma}^{-1/2}(X_i - \bar{X}) \quad \text{for } i = 1, \dots, n$$

$$\hat{g}_h(y) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h^d} K\left(\frac{y - Y_i}{h}\right)$$

$$g(y) : \text{density of } Y, \quad Y = A(x - b)$$

$$f(x) = g(y) \Big|_{y \rightarrow x} |J| =$$

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h^d} |\hat{\Sigma}|^{-1/2} K\left(\frac{\hat{\Sigma}^{-1/2}(x - X_i)}{h}\right)$$

where

$$y = \hat{\Sigma}^{-1/2}(x - \bar{X}), \quad Y_i = \hat{\Sigma}^{-1/2}(X_i - \bar{X})$$

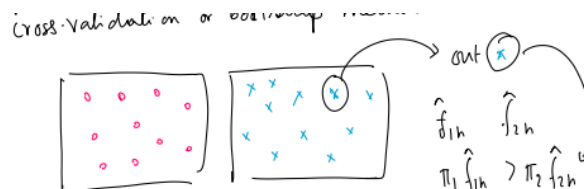
KDA Classifier

or

$$\hat{\delta}_{\text{KDA}}(x) = \underset{1 \leq i \leq J}{\operatorname{argmax}} \pi_i \hat{f}_{i,h}(x)$$

$$\underset{1 \leq i \leq J}{\operatorname{argmax}} \pi_i \hat{f}_{i,h_i}(x)$$

$$\pi_1 \hat{f}_1(x) > \pi_2 \hat{f}_2(x)$$



KDA classification diagram: two class groups (circles and crosses) with a query point x to be classified. Estimate $\hat{f}_{1h}$ and $\hat{f}_{2h}$; classify by $\pi_1 \hat{f}_{1h} > \pi_2 \hat{f}_{2h}$.

Bandwidth Selection for KDA

In the case of KDA, we do **not** choose the bandwidth using the least square cross validation method.

In the case of classification, bandwidths are usually chosen by **cross-validation** or **bootstrap method**.

Summary:

- Fit $\hat{f}_{1h}$ and $\hat{f}_{2h}$ on training data.
- For an out-of-sample x : classify by $\pi_1 \hat{f}_{1h} > \pi_2 \hat{f}_{2h}$.

End of Notes